

Hankel-Structured Channel Estimation for Magnitude-Only Rydberg Atomic Receivers: When Does Structure Help?

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Abstract—We study channel estimation for a magnitude-only Rydberg atomic receiver array, in which the optical readout provides the amplitude of the incident radio-frequency field but not its phase, so complex channel estimation becomes a phase-retrieval problem. Rydberg architectures that recover phase, such as a superheterodyne readout, are outside this scope. This paper asks when the known spatial structure of a multipath channel improves such magnitude-only estimation, and when that structure is too restrictive to help. We propose a Hankel-structured extension of expectation-maximization Gerchberg-Saxton (EM-GS) that exploits the relationship between channel coefficients across a uniform linear array, choosing the amount of structure from held-out pilot measurements. Selecting the full rank disables the structural step, and the estimator then returns the EM-GS answer exactly; this is a property of the operator and not a guarantee that the selection rule avoids harmful choices. Simulations show that the structural step helps when the channel is simple relative to the structural capacity of the array, and that its benefit falls, and eventually reverses, as channel complexity approaches that capacity. In a held-out comparison, path count normalized by that capacity predicts the measured gain far better than path count alone, and the prediction transfers to two channel distributions not used to build it. A constrained Cramér-Rao bound for the multipath model, derived and verified here, quantifies what the structure is worth in principle. These results give a practical account of when spatial structure should be exploited in magnitude-only channel estimation and when an estimator should rely on the unconstrained solution instead.

Index Terms—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, constrained Cramér-Rao bound, effective rank.

I. INTRODUCTION

SIXTH-GENERATION wireless systems are expected to operate over wider bandwidths and at increasingly high carrier frequencies, placing new demands on receiver sensitivity and reconfigurability. Quantum sensing has emerged as one possible response, and Rydberg atomic receivers have recently attracted attention for wireless communications [1], [2]. These receivers use laser-excited atoms to sense an incident radio-frequency field through changes in optical transmission. Unlike a conventional coherent receiver, however, the optical readout directly provides the field amplitude rather than its complex value, so the channel phase is not directly observed.

Recovering a complex channel from these amplitude-only measurements is a phase-retrieval problem. A known reference field makes this recovery possible by converting relative phase

into observable amplitude. Existing approaches include iterative phase-retrieval methods [3], gradient descent with structural projection [4], and unrolled networks [5]. These works focus mainly on improving channel estimation performance. Less attention has been given to a more basic question: when does imposing channel structure actually help, and when does that structure become too restrictive? This paper addresses that question using the spatial structure of multipath channels on a uniform linear array.

We propose a Hankel-structured extension of EM-GS for magnitude-only channel estimation. The estimator exploits the exponential structure of the channel across the array, selects the structural order using held-out pilot measurements. When the selection returns the full rank the structural step changes nothing and the estimator reproduces EM-GS exactly; this is a property of the operator, and it is not a guarantee that the selection rule avoids choices that hurt. We then study how the benefit of the Hankel constraint changes with the array size and the complexity of the channel. Path count alone does not fully describe channel complexity, so we also describe channel complexity using its effective rank relative to the available Hankel rank. Finally, we test this relationship on two additional channel distributions.

II. SYSTEM MODEL

We consider an uplink system with K single-antenna users and one Rydberg atomic receiver, as shown in Fig. 1. The receiver contains N vapour-cell receive elements arranged in a uniform linear array (ULA). Each vapour cell contains atoms that are excited to Rydberg states using probe and coupling lasers. The RF signals transmitted by the users interact with these atoms. A known RF reference field is also applied at the receiver. The atomic response changes the probe laser passing through the vapour cell, and this change is measured using a photodetector. The receiver therefore measures the strength of the total RF field, but it does not directly observe its complex phase [1].

This paper studies one Rydberg architecture, not all of them. There are Rydberg receivers that use a superheterodyne readout, in which a local reference drives the vapour cell and the beat note is detected. Those receivers recover both amplitude and phase, so the problem studied here does not arise for them. Everything below applies to the magnitude-only readout described above. In fact, nothing below depends on the atomic physics itself: the results apply to any array

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receiver that measures only the magnitude of the total field, and the Rydberg receiver is the motivating example.

We model the channel between each user and the array using several propagation paths. A path may represent a signal arriving directly from the user or after interacting with objects in the environment. For user k , the channel coefficient at the n th receive element is

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \quad \psi_{\ell,k} = \pi \sin \theta_{\ell,k}, \quad (1)$$

where L_k is the number of propagation paths for user k , $\alpha_{\ell,k}$ is the complex gain of path ℓ , and $\theta_{\ell,k}$ is its angle of arrival. In our simulations, L_k is selected between 3 and 7, following the channel setting in [4].

The exponential term in (1) describes how the phase of one propagation path changes as we move from one array element to the next. To see this, consider a single path arriving at the array from an angle θ . The signal reaches the different array elements at slightly different times because the elements are at different positions. This difference in travel distance produces a phase difference between adjacent elements.

With half-wavelength spacing between the array elements, the phase change between two neighbouring elements is

$$\psi = \pi \sin \theta. \quad (2)$$

The first array element has no additional phase shift from the array position. The second element has a phase shift of ψ , the third has a phase shift of 2ψ , and so on. Therefore, the samples produced by one path across the array have the form

$$1, \quad e^{-j\psi}, \quad e^{-j2\psi}, \quad \dots, \quad e^{-j(N-1)\psi}. \quad (3)$$

The factor

$$e^{-j(n-1)\psi_{\ell,k}} \quad (4)$$

in (1) represents exactly this phase pattern for path ℓ . The coefficient $\alpha_{\ell,k}$ determines the strength and initial phase of that path. Since several paths can arrive at the receiver, their contributions are added together to produce the final channel coefficient $g_k[n]$.

The channel coefficients of user k are collected into the vector

$$\mathbf{g}_k = [g_k[1] \quad g_k[2] \quad \dots \quad g_k[N]]^T, \quad (5)$$

and the channels of all K users are collected into

$$\mathbf{G} = [\mathbf{g}_1 \quad \mathbf{g}_2 \quad \dots \quad \mathbf{g}_K] \in \mathbb{C}^{N \times K}. \quad (6)$$

During channel estimation, the K users transmit P known pilot symbols. These pilots are collected in

$$\mathbf{S} \in \mathbb{C}^{K \times P}. \quad (7)$$

Each row of $\mathbf{S}$ contains the pilots transmitted by one user, while each column represents one pilot transmission. The same pilot matrix is used when comparing the different channel estimators.

If the receiver could observe the complete complex signal, the contribution from the users would be $\mathbf{GS}$. However, the Rydberg receiver considered here measures magnitude rather than directly measuring this complex value. A known reference

RF field is therefore applied at the receiver. We denote this reference by

$$\mathbf{B} \in \mathbb{C}^{N \times P}. \quad (8)$$

The reference field is known in both amplitude and phase. It is important because the magnitude of an unknown signal alone does not directly reveal its phase. When the unknown user signal is combined with a known reference, the resulting magnitude depends on their relative phase. This provides information that can be used to estimate the complex channel.

Including noise, the field before the magnitude measurement is

$$\mathbf{Y} = \mathbf{GS} + \mathbf{B} + \mathbf{W}, \quad (9)$$

where the noise is modeled as complex Gaussian,

$$\mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2). \quad (10)$$

The noise is therefore added before the magnitude is measured. The Rydberg receiver observes

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \quad (11)$$

where $|\cdot|$ is applied to every element of the matrix.

Equation (11) is the main measurement model used in this work. The pilots $\mathbf{S}$ and reference field $\mathbf{B}$ are known, and the receiver measures $\mathbf{Z}$. The unknown quantity is the complex channel matrix $\mathbf{G}$. The channel-estimation problem is therefore to recover $\mathbf{G}$ even though the receiver does not directly observe the phase of the received signal.

We assume a narrowband, frequency-flat channel during the pilot interval. The path gains are modeled as independent circularly symmetric complex Gaussian variables, and the angles of arrival are independently generated over $(-90^\circ, 90^\circ)$. The users have equal received power, and the reference field remains known at the receiver. For a fair comparison, all estimators use the same channel, pilots, reference field, and noise realization within each Monte Carlo trial.

The path gains are scaled so that the total received power does not change with the number of paths. We draw

$$\alpha_{\ell,k} \sim \mathcal{CN}(0, \beta_k/L_k), \quad (12)$$

so that the powers of the individual paths add up to

$$\sum_{\ell=1}^{L_k} \mathbb{E} |\alpha_{\ell,k}|^2 = \beta_k, \quad (13)$$

which does not depend on L_k . This matters wherever the number of paths is varied. Without this scaling, adding more paths would also add more received power, and we could not tell whether a change in performance came from the extra complexity or simply from the extra power. With this scaling, only the complexity changes.

The reference-to-signal ratio (RSR) is the reference power divided by the single-user received signal power. The signal-to-noise ratio (SNR) is the single-user received signal power divided by σ^2 .

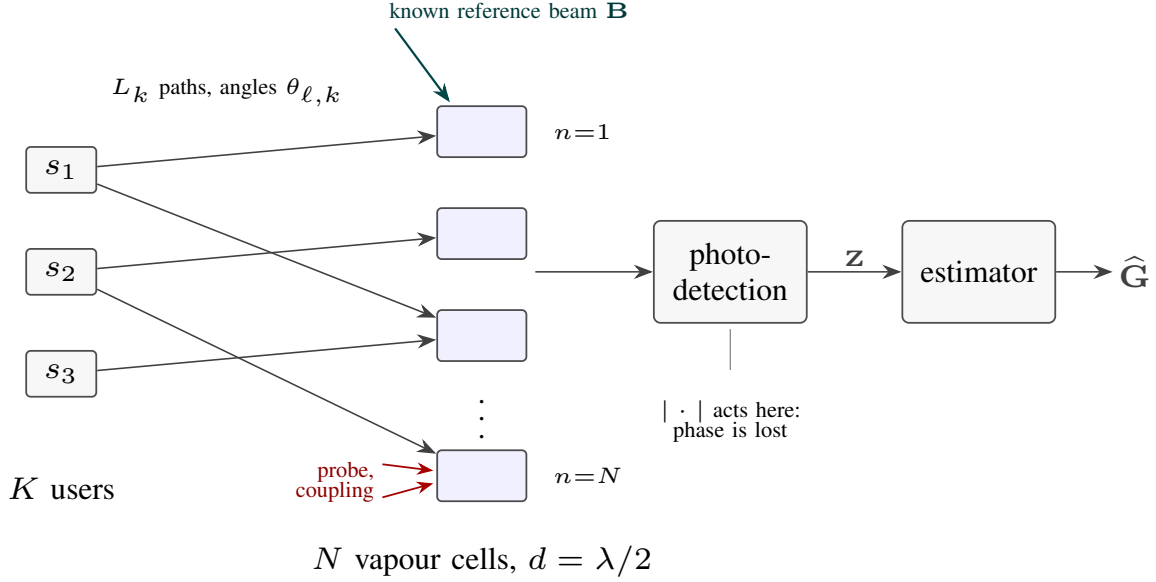


Fig. 1. Uplink of the Rydberg atomic receiver. K single-antenna users transmit known pilot signals to N vapour-cell receive elements arranged as a uniform linear array. The user RF signals and a known reference RF field interact with the Rydberg atoms. Probe and coupling lasers are used for the atomic readout, and the photodetector measures the magnitude of the total field. The estimator therefore observes $\mathbf{Z}$ without directly observing its phase.

III. PROPOSED ESTIMATOR

A. Structure and its Capacity

The rank of a matrix tells us how many independent components are needed to describe it. A low-rank matrix is therefore described using only a small number of important components. To use this idea for our channel, we first rewrite the channel into a Hankel matrix. A Hankel matrix is a matrix in which the same channel samples are repeated along the antidiagonals. For example, consider

$$\mathbf{g} = [g_1 \ g_2 \ g_3 \ g_4 \ g_5]^T. \quad (14)$$

One possible Hankel representation is

$$\mathcal{H}(\mathbf{g}) = \begin{bmatrix} g_1 & g_2 & g_3 \\ g_2 & g_3 & g_4 \\ g_3 & g_4 & g_5 \end{bmatrix}. \quad (15)$$

We use this representation because the channel in (1) is not a random collection of values. It is made up of propagation paths, and each path produces a complex exponential pattern across the array elements. When this type of channel is written as a Hankel matrix, each path contributes one rank-one component. For example, a channel containing one path produces a Hankel matrix with rank at most one. Two paths produce a matrix with rank at most two. In general, for a channel with L_k paths,

$$\text{rank } \mathcal{H}_p(\mathbf{g}_k) \leq L_k. \quad (16)$$

Therefore, when the channel has only a few paths, its Hankel matrix can have a low rank [11], [12]. This is why we use the Hankel representation: it changes the multipath structure of the channel into a low-rank matrix structure that can be used during channel estimation.

One point of care. The bound is on the number of paths, not on the number of *strong* paths. A channel with three strong

paths and five weak ones has Hankel rank at most 8, not 3; the weak paths still contribute rank-one components, however small their singular values. Such a channel is *approximately* low rank rather than exactly low rank. The distinction matters, because a rank bound counts paths while the usefulness of the constraint depends on how many of them carry real energy.

The maximum possible rank of the Hankel matrix depends on the number of array elements. For an $(N - p) \times (p + 1)$ Hankel matrix, the maximum rank is

$$\min(N - p, p + 1). \quad (17)$$

Choosing p so that the matrix is as balanced as possible gives

$$r_{\max}(N) = \left\lceil \frac{N}{2} \right\rceil. \quad (18)$$

For example, if $N = 16$, the maximum Hankel rank is 8. If the channel has only three paths, its Hankel rank can be at most 3. This is much smaller than the maximum rank of 8, so there is useful low-rank structure that can be used during estimation. If the channel rank is already close to 8, there is less low-rank structure available. In this case, forcing the channel to have a much smaller rank may remove part of the real channel.

B. The Baseline: EM-GS

Our starting point is the expectation-maximization Gerchberg-Saxton estimator, which we call EM-GS, given by Cui *et al.* [1, Alg. 2] as an adaptation of the classical Gerchberg-Saxton iteration [3] to this receiver. Because the whole paper is a comparison against it, it is worth writing down.

The difficulty is that we measure $\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|$ and the phase is missing. If we knew the phase, the problem would

be ordinary linear estimation. EM-GS therefore alternates between guessing the phase and solving the linear problem that the guess implies.

Let $\hat{\mathbf{G}}^{(t)}$ be the channel estimate after t iterations, and let $\mathbf{\Lambda}^{(t)} = \hat{\mathbf{G}}^{(t)}\mathbf{S} + \mathbf{B}$ be the noiseless field it predicts. One iteration is two steps.

Expectation step. Give every measured magnitude the phase that the current estimate predicts for it, producing a complex field

$$\mathbf{Y}_{np}^{(t)} = \mathbf{Z}_{np} R(\kappa_{np}) \frac{\mathbf{\Lambda}_{np}^{(t)}}{|\mathbf{\Lambda}_{np}^{(t)}|}, \quad \kappa_{np} = \frac{2\mathbf{Z}_{np}|\mathbf{\Lambda}_{np}^{(t)}|}{\sigma^2}, \quad (19)$$

where $R(x) = I_1(x)/I_0(x)$ is the same Bessel ratio that appears in the Rician model. This is the conditional mean of the noiseless field given the measurement and the current estimate, which is what makes the step an expectation step rather than a plain phase substitution. Because $R(x) < 1$, the imputed field is *shrunk* towards zero relative to the measurement, and the shrinkage is strongest where the measurement is least informative. At high SNR $R \rightarrow 1$ and (19) reduces to substituting the predicted phase and keeping the measured magnitude.

Maximization step. Treat $\mathbf{Y}^{(t)}$ as if it were an observed complex measurement and solve the resulting least-squares problem,

$$\hat{\mathbf{G}}^{(t+1)} = (\mathbf{Y}^{(t)} - \mathbf{B})\mathbf{S}^H (\mathbf{S}\mathbf{S}^H + \varepsilon\mathbf{I})^{-1}, \quad (20)$$

with a ridge ε that is set to zero in all experiments here and is retained only to guard against a singular pilot Gram matrix.

The iteration is started from the spectral initializer of [1], computed independently for each receive element, and not from zero. It then runs for a fixed budget of T iterations rather than to a convergence tolerance, with T fixed per configuration and stated with the simulation settings. A fixed budget is used because it makes the comparison between estimators exact: both arms get the same number of iterations, so no difference between them can come from one being allowed to run longer. It also means neither arm is claimed to have converged.

Nothing in (19) or (20) knows that the channel comes from a small number of propagation paths. Each channel coefficient is treated as a free complex number.

C. The Estimator

EM-GS does not follow the rank structure discussed earlier. To include that structure, we apply a Cadzow denoising step to the channel estimate. There are three parts to the step. First, the channel estimate is converted into the Hankel matrix described earlier.

Next, singular value decomposition (SVD) is applied to the Hankel matrix. SVD separates the matrix into components and gives a singular value for each component. A larger singular value means that the component has a larger contribution to the matrix. For example, suppose the singular values are

$$\sigma_1 = 10, \quad \sigma_2 = 6, \quad \sigma_3 = 0.3. \quad (21)$$

Here, the three singular values are 10, 6, and 0.3. If we decide to use a rank of 2, we keep the two largest components,

corresponding to 10 and 6, and remove the smaller component corresponding to 0.3. The resulting matrix therefore has rank 2. If the true channel has only two important components, the removed component may mainly come from noise or estimation error.

After the rank reduction, the matrix is converted back into a channel vector. The entries along each antidiagonal are averaged because they represent the same channel sample. These steps can be written as

$$\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger (\Pi_r (\mathcal{H}_p(\mathbf{g}))), \quad (22)$$

where Π_r keeps the r largest singular values and $\mathcal{H}_p^\dagger$ converts the Hankel matrix back into a channel vector by averaging its antidiagonals.

This step is a denoising step, not a projection. It is worth being careful about the word, because the two are not the same thing. A projection must land on the set it is projecting onto, and applying it a second time must change nothing. The map in (22) does neither. Rank reduction generally destroys the Hankel pattern, and averaging the antidiagonals to restore that pattern generally raises the rank again.

A small example makes this concrete. Take

$$\mathbf{g} = [1 \quad 2 \quad 3 \quad 4 \quad 5]^T \quad (23)$$

and use the 3×3 Hankel matrix above. Applying (22) once with $r = 1$ and forming the Hankel matrix of the result gives singular values

$$[9.6194, \quad 0.1295, \quad 0.0616], \quad (24)$$

which is a matrix of rank 3, not rank 1. Applying the same step a second time moves the answer again, by about 0.099 in Euclidean distance. A projection would not have moved it at all. We therefore call (22) a Cadzow step [11], and we apply it repeatedly rather than once, so that the number of sweeps is part of the definition of the operator. Even when repeated sweeps settle down, they are not guaranteed to reach the closest matrix that is both Hankel and low rank [12].

A low Hankel rank is also a weaker requirement than our channel model. Equation (1) says the channel is a sum of exponentials whose modes lie on the unit circle, because $e^{-j(n-1)\psi}$ always has magnitude one. A low-rank Hankel matrix only says the channel is a sum of a few exponentials; their modes may grow or decay across the array and need not be angles of arrival at all. Imposing low Hankel rank is therefore a relaxation of the array model, not the array model itself. We use it because it is easy to impose and because the true channel does satisfy it, but the constraint is looser than the physics.

We call the resulting method Hankel-structured EM-GS (HS-GS).

Where the step is applied, precisely. HS-GS applies the Cadzow step (22) to each column of the channel estimate *after every one of the T EM-GS updates*, not once at the end. So a single HS-GS iteration is the expectation step (19), the least-squares step (20), and then the structural step, in that order, with the result of the structural step becoming the estimate the next expectation step predicts from. This matters

because the two orderings do not produce the same answer: denoising an estimate and then iterating is not the same operation as iterating and then denoising once, and neither is a special case of the other. The alternative — running EM-GS to its full budget and applying one Cadzow step to the result — is a different and cheaper estimator, and it is measured separately among the numerical results rather than treated as interchangeable with this one.

Choosing r is important because a rank that is too large may remove very little, while a rank that is too small may remove useful parts of the channel.

D. Choosing the Hankel Rank

The Cadzow step requires us to choose the rank r . However, the correct rank is not known at the receiver, so we choose it using the pilot measurements.

The selection is done in two stages, and it is worth separating them clearly.

Stage one, choosing the rank. We set aside 30% of the pilot columns. For each possible rank,

$$r \in \{1, 2, \dots, r_{\max}\}, \quad (25)$$

we estimate the channel using only the other 70%, and then use that estimate to predict the magnitude measurements of the held-out pilots. We compare these predicted magnitudes with the actual measurements. The rank with the smallest difference, or residual, is selected. For example, if 30 pilots are available, about 21 are used for these trial estimates and 9 are kept for scoring them.

Stage two, the final estimate. Once the rank $\hat{r}$ has been chosen, the estimate is computed again using *all* P pilot columns, not only the 70%. The split is used for scoring, and nothing else.

This distinction matters for a fair comparison. HS-GS and EM-GS therefore use exactly the same number of pilots to produce their final estimates, and the difference between them is only the Cadzow step. It also means that when the step is switched off, HS-GS returns exactly the EM-GS answer: in our simulations the two agree to $\max |\text{diff}| = 0$, so the Cadzow step is provably the only difference between them.

We cannot score the candidate ranks using the pilots that were used to fit them. Constraining the channel can only make the fit to those pilots worse, so an in-sample score would always prefer the largest rank, which is the same as using no constraint at all. Scoring on pilots that were held out penalises over-modelling correctly.

For example, suppose $r_{\max} = 8$. We test ranks from 1 to 8. If $r = 3$ gives the smallest error on the held-out pilots, we choose

$$\hat{r} = 3. \quad (26)$$

We then keep the three strongest singular-value components in the Cadzow step.

This method does not require the true number of propagation paths L_k . If the selected rank is $r_{\max}$, all singular-value components are kept, the Cadzow step does not lower the rank, and HS-GS returns exactly the EM-GS estimate.

It is worth being exact about what that does and does not promise. It says the estimator has a harmless setting available, and that selecting it costs nothing. It does *not* say the selection rule will choose that setting whenever the constraint would hurt. The rule scores candidate ranks on held-out pilots, and a rank that scores well there can still damage the estimate, so a selected rank below $r_{\max}$ is not evidence that constraining was the right choice on that trial.

IV. PERFORMANCE REFERENCES

A. Information in an Amplitude Measurement

The Cramér–Rao lower bound (CRLB) is used as a theoretical reference for channel estimation. In simple terms, it tells us how accurately an unbiased estimator could estimate the channel from the available measurements. The amount of information contained in the measurements therefore affects the CRLB.

This is important for the Rydberg receiver because it measures only the amplitude of the received signal, not its full complex value. For example, suppose the noiseless received signal is

$$s = 3 + j4. \quad (27)$$

A complex receiver can observe both the real part, 3, and the imaginary part, 4. The amplitude-only receiver instead observes

$$|s| = \sqrt{3^2 + 4^2} = 5. \quad (28)$$

Knowing only the value 5 does not tell us that the original signal was $3 + j4$. Other complex values can have the same amplitude. Therefore, an amplitude measurement contains less information than a full complex measurement.

When complex Gaussian noise is added before the amplitude is measured, the measured amplitude follows a Rician distribution. We use this Rician model to calculate the Fisher information matrix J and the corresponding CRLB [6], [1]. The Fisher information matrix describes how much information the measurements contain about the unknown channel.

A simple example can be used to understand J . Let the complex signal be

$$s = x + jy, \quad (29)$$

where x is the real part and y is the imaginary part. The receiver measures only

$$a = |s| = \sqrt{x^2 + y^2}. \quad (30)$$

For example, if $x = 3$ and $y = 4$, then $a = 5$. We can then check how much the measured amplitude changes when x or y changes:

$$\begin{bmatrix} \frac{\partial a}{\partial x} \\ \frac{\partial a}{\partial y} \end{bmatrix} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}. \quad (31)$$

This tells us the direction in which this amplitude measurement provides information about the unknown complex signal. The Fisher information from this measurement has the form

$$J \propto \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} \begin{bmatrix} 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix}. \quad (32)$$

This matrix has rank one. This means that one amplitude measurement gives information in only one direction. We do not have enough information from this one measurement to find the real and imaginary parts separately. This agrees with the earlier example: knowing that the amplitude is 5 is not enough to know that the signal is exactly $3 + j4$.

In practice, we use several pilot measurements. Each pilot gives more information about the unknown channel. The information from all of these measurements is combined to form the full Fisher information matrix J . Therefore, J describes the information provided by all the measurements used for channel estimation.

One feature of J is worth stating plainly, because it explains what a larger array does. Measurement (n, p) depends only on row n of $\mathbf{GS}$, because row n of the array produces row n of the measurements and nothing else. So J is *block diagonal across the receive elements*: there is one $2K \times 2K$ block per element and no coupling between them. Adding array elements therefore adds measurements and unknown channel coefficients in a fixed proportion. An estimator that treats the channel coefficients as unrelated gains nothing from a larger array under a normalised error measure, because the extra measurements are used up by the extra unknowns.

The Hankel structure breaks that proportion. The channel coefficients across the array are not really independent, because they are produced by the same propagation paths. The Hankel constraint uses this relationship between the array elements, so the number of quantities actually being estimated does not grow with N in the same way. This is what allows HS-GS to make use of the additional spatial structure provided by a larger array.

B. Constrained Cramér–Rao Bound

The CRLB described above treats each channel coefficient as a separate unknown. However, our channel coefficients are connected because they are produced by the same propagation paths. For example, consider

$$\mathbf{g} = [g_1 \ g_2 \ g_3 \ g_4]^T. \quad (33)$$

The ordinary CRLB allows g_1, g_2, g_3 , and g_4 to take any values. In our channel model, these values are not completely independent. They are connected through the path gains and angles in (1). The constrained CRLB (CCRB) includes this additional information.

The rest of this subsection gives the bound in enough detail to be recomputed. It is written out step by step because a reference curve nobody can redraw is not a reference.

1) *The parameters we differentiate with respect to:* The CCRB is a bound for estimators that know the channel follows (1). So the unknowns are not the channel coefficients. They are the path parameters that generate them. For each user k and each path ℓ we treat three real numbers as unknown: the spatial frequency $\psi_{\ell,k}$, and the real and imaginary parts of the path gain $\alpha_{\ell,k}$. Collecting them gives

$$\phi = \{\psi_{\ell,k}, \text{Re } \alpha_{\ell,k}, \text{Im } \alpha_{\ell,k}\}, \quad m = 3 \sum_{k=1}^K L_k. \quad (34)$$

Two things about this are worth saying out loud. First, the angular unknown is the spatial frequency $\psi = \pi \sin \theta$ and not the arrival angle θ , because ψ is the variable that actually appears in (1). Differentiating with respect to θ instead evaluates the bound at the wrong point and gives a different, wrong curve. Second, the number of paths L_k is treated as *known*. The bound is therefore a local bound for a receiver that already knows how many paths there are, which is more than our estimator knows.

A worked count, at the setting used for our main figure. With $K = 3$ users and L_k drawn between 3 and 7, the average is $L_k = 5$ per user, so

$$m = 3 \times (5 + 5 + 5) = 45. \quad (35)$$

The channel itself has $2NK = 2 \times 32 \times 3 = 192$ real numbers in it. So the model says those 192 numbers are really controlled by about 45. That reduction is the whole content of the constraint.

2) *How the channel changes when a parameter changes:* Write the channel in real coordinates: for each element n , list the K real parts and then the K imaginary parts, and stack the N groups. That is a real vector of length $2NK$. Let D be the matrix of derivatives of that vector with respect to ϕ . Its three kinds of column come straight from differentiating (1),

$$\begin{aligned} \frac{\partial g_k[n]}{\partial \psi_{\ell,k}} &= -j(n-1) \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \\ \frac{\partial g_k[n]}{\partial \text{Re } \alpha_{\ell,k}} &= e^{-j(n-1)\psi_{\ell,k}}, \quad \frac{\partial g_k[n]}{\partial \text{Im } \alpha_{\ell,k}} = j e^{-j(n-1)\psi_{\ell,k}}. \end{aligned} \quad (36)$$

Each complex derivative contributes its real part to one row of D and its imaginary part to another, matching the ordering above.

3) *The information matrix:* The measured amplitude is Rician, with the density

$$p(z | \lambda) = \frac{2z}{\sigma^2} \exp\left(-\frac{z^2 + |\lambda|^2}{\sigma^2}\right) I_0\left(\frac{2z|\lambda|}{\sigma^2}\right), \quad z \geq 0, \quad (37)$$

where $\lambda = \mathbf{GS} + \mathbf{B}$ is the noiseless field and I_0 is the modified Bessel function of the first kind. The density depends on the parameters only through $a = |\lambda|$, so one measurement carries a single scalar amount of information,

$$I(a) = 4\beta(a, \sigma^2), \quad \beta = \frac{\mathbb{E}[z^2 R^2(\kappa)] - a^2}{\sigma^4}, \quad (38)$$

with $\kappa = 2za/\sigma^2$ and $R(x) = I_1(x)/I_0(x)$. The expectation is taken under (37) and computed by numerical integration.

That scalar is turned into a matrix by the direction in which the measurement is informative. Writing $c_k = e^{-j\angle \lambda_{n,p}} S_{k,p}$ and $\mathbf{u}_{n,p} = [\text{Re } c_1 \dots \text{Re } c_K, -\text{Im } c_1 \dots -\text{Im } c_K]^T$, the block of J for element n is

$$J_n = \sum_{p=1}^P 4\beta(a_{n,p}, \sigma^2) \mathbf{u}_{n,p} \mathbf{u}_{n,p}^T. \quad (39)$$

Every term is an outer product of one vector with itself, so every term has rank one. This is the same point made by the $3 + j4$ example above, now written for the full problem: a magnitude measurement informs one real direction, not two.

4) Why the bound uses a basis and not the derivatives:

The textbook constrained bound is $D(D^T J D)^{-1} D^T$. We do not use that form, because $D^T J D$ is often singular here. The reason is that the path description is over-complete: if m in (34) exceeds the $2NK$ real numbers in the channel, different parameter sets produce the same channel and the derivatives cannot be independent. This is not a rare corner. With $K = 3$ and L_k between 3 and 7, it happens in exactly 28.0% of trials at $N = 8$, where the channel holds only $2NK = 48$ real numbers. At $N = 16$ and $N = 32$ it never happens, because $2NK$ is 96 and 192 against at most $3 \times 21 = 63$ parameters.

The bound itself is still well defined, because it depends only on which directions the channel can move in — the span of the columns of D — and not on the parameters being separately identifiable. So we take U to be an orthonormal basis of that span, obtained from the singular value decomposition of D and keeping the singular values above 10^{-9} times the largest. Then

$$\text{CCRB} = U(U^T J U)^{-1} U^T. \quad (40)$$

This form is invariant to how the paths are parameterised, which the textbook form is not. The measured number of kept directions ranges from 40.4 to 44.6 across our operating points, against the 45 parameters an average trial has, and the matrix $U^T J U$ that gets inverted has a condition number of about 3.1 at the main configuration and never worse than 52 anywhere. The inverse in (40) is therefore not a delicate one.

5) *Turning the bound into an NMSE:* Equation (40) bounds a covariance, and the figures plot a normalised error. The trace of the CCRB bounds the expected squared error of the whole channel matrix, so we report

$$\text{NMSE}_{\text{CCRB}} = 10 \log_{10} \frac{\sum_t \text{Tr}(\text{CCRB}_t)}{\sum_t \|\mathbf{G}_t\|_F^2}, \quad (41)$$

where t runs over trials. Note that this is a ratio of two sums, not an average of per-trial ratios; the two are not the same and we use the first. The trials are the same 400 channel realisations the estimator curves use, so the bound and the measured curves are paired on the channel draws rather than averaged over different ones. The bound does not depend on the noise realisation, only on $\mathbf{G}$, $\mathbf{S}$, $\mathbf{B}$ and σ^2 .

6) *Checking the implementation:* Two checks are run before the bound is computed, and both are arithmetic identities rather than opinions. The first compares our real-coordinate bound against the convention that credits both quadratures of each measurement. Crediting both can only add information, so our bound must come out strictly higher. It does, by 0.73 to 2.88 dB over four test cells, paired on the same trials.

The second is a high-SNR limit. As the noise falls, $\beta \rightarrow 1/(2\sigma^2)$, so the amplitude-only bound must approach exactly $10 \log_{10} 2 = 3.0103$ dB above the error of a receiver that is told the true phase. Measured at 45 dB SNR, the gap is 3.55 dB at $N = 8$ and 3.52 dB at $N = 16$. These sit about half a decibel above the limit because at a finite pilot count the rank-one terms in (39) have not yet averaged out; the agreement is a limit, not an identity at finite P , and we report the measured numbers rather than the limit alone.

One measured weakness belongs here rather than in a footnote. To reach 400 trials per point, β is read from an interpolation table instead of being integrated for every measurement, and the error of that table is measured at run time. Across the points plotted in this paper the largest relative error is 7.4×10^{-4} . Two points computed for a reference-power sweep that this paper does not plot are much worse, at 1.9×10^{-2} and 5.0×10^{-2} , both at a reference-to-signal ratio of 0 dB where the amplitude distribution is widest.

7) *What the bounds do and do not govern:* The ordinary CRLB treats the channel coefficients separately, while the CCRB uses the known multipath structure. The difference between them is what the structural model is worth in principle.

Both bounds constrain unbiased estimators. EM-GS and HS-GS are not unbiased: both run to a fixed iteration budget, and HS-GS additionally truncates rank. So neither bound applies strictly, and we do not treat either as an absolute limit on the measured NMSE. The CCRB is the reference matched to the model class that HS-GS assumes; the unconstrained CRLB governs EM-GS only.

A last distinction that is easy to blur. The CCRB is a bound for the physical multipath model of (1). It is not a bound for a fixed-rank Hankel model, which is a looser description admitting channels that (1) excludes. The two should not be used interchangeably.

V. NUMERICAL RESULTS

A. Experimental Setup

We use simulations to compare EM-GS with the proposed HS-GS estimator. The channels are generated using the multipath model in (1). Unless otherwise stated, we use $N = 32$ array elements, $K = 3$ users, and an RSR of 12 dB.

Each result is obtained using 300–400 Monte Carlo trials. In simple terms, this means that we repeat the simulation hundreds of times with different channel and noise realizations. For a fair comparison, EM-GS and HS-GS use exactly the same data in each trial. For example, if one trial generates a particular channel, pilot sequence, reference signal, and noise realization, both estimators receive those same values. Therefore, any difference between their results comes from the Cadzow step used by HS-GS. If the step is disabled, HS-GS gives the same result as EM-GS.

The two generators. TB draws L_k between 3 and 7 independently per user, with gains scaled as in (12) and angles uniform on $(-90^\circ, 90^\circ)$. TD is the generator used by the unrolled-network experiments this work sits alongside; it follows the same model (1) but fixes L per cell rather than drawing it, which is what makes a path-count sweep possible. Both stay inside the geometric ULA family.

How error is measured. For one trial, the normalized error of an estimate is

$$\text{NMSE} = \frac{\|\hat{\mathbf{G}} - \mathbf{G}\|_F^2}{\|\mathbf{G}\|_F^2}, \quad (42)$$

and the per-trial gain in decibels is the difference of the two arms' errors on *that same trial*,

$$\Delta_{\text{HS}}^{(i)} = 10 \log_{10} \text{NMSE}_{\text{EM-GS}}^{(i)} - 10 \log_{10} \text{NMSE}_{\text{HS-GS}}^{(i)}. \quad (43)$$

Positive means HS-GS is better on that trial.

Three summaries, and they are not interchangeable. The *paired median* is the median of (43) over trials: every trial counts once, whatever its difficulty. The *mean dB gain* averages (43) instead, so a few very good or very bad trials move it. The *ratio of summed errors* adds the squared errors of both arms across trials and compares the totals, which weights trials by how hard they are and is therefore dominated by the worst ones. The paired median is our primary summary throughout; where the three disagree we say so and report more than one, because a disagreement between them is information about the distribution rather than a detail of presentation.

Aggregating operating points. The array-size experiment has two pilot counts and six SNR values at each N , so twelve operating points. We report the unweighted mean of the twelve per-point summaries, not a pooled statistic over the trials of all twelve: pooling would weight the points by trial count and by how noisy each point is, which is not what a sweep over conditions is meant to show.

Trial counts. Most cells use 300 to 400 paired trials. Two cells do not: the cross-generator tests use 276 and 266. Those two counts are what the stored result files contain, and we report them as measured. We have not established the cause of the shortfall and do not assert one here.

Predictions registered before the runs. Each experiment in this section was preceded by a dated record fixing what we expected to see, the threshold that would count as failure, and the rule that would decide. The records were written before the corresponding runs and scored afterwards. Four such records govern the results in this paper, and *two of the four failed*:

- the predicted improvement of HS-GS over EM-GS at the main configuration, a six-point mean in [2.2, 2.9] dB: **held**, measured 2.68 dB;
- the predicted array-size means, one band per N : **failed**. The $N = 16$ mean came out at 0.966 dB against a predicted [0.60, 0.95], missing the band by 0.016 dB, and at $N = 8$ three of the twelve operating points were positive with intervals excluding zero, where we had predicted the method would not help;
- the predicted effect of the Hankel pencil choice, a span below 0.30 dB: **failed**, measured 0.391 dB;
- the predicted advantage of applying the structural step during the iteration rather than once at the end, predicted at [0.3, 1.5] dB: **failed**, measured 0.003 dB.

We report these because a paper that registers predictions and then mentions only the ones that held has gained nothing from registering them. Each failure is discussed where its experiment is presented, rather than collected out of sight here.

Table I summarizes the four simulation configurations used in the experiments.

B. Performance Against the Reference Curves

We first compare EM-GS and HS-GS with the CRLB and CCRB discussed in Section IV. Fig. 2(a) shows the NMSE as the SNR changes from -5 to 20 dB. This experiment uses configuration A with $N = 32$ and $P = 30$.

TABLE I

SIMULATION CONFIGURATIONS USED IN THE NUMERICAL EXPERIMENTS. ALL FOUR USE THE SAME ESTIMATOR AND THE SAME HELD-OUT ORDER RULE (CANDIDATES $1 \dots r_{\max}$, 30% OF PILOT COLUMNS HELD OUT FOR SCORING) AND ALL FOUR USE $n_{cz} = 4$ CADZOW SWEEPS PER APPLICATION OF THE STRUCTURAL STEP; THEY DIFFER ONLY IN OPERATING POINT. T IS THE EM-GS ITERATION BUDGET AND “ORDER SEARCH” THE ITERATION BUDGET USED WHILE SCORING CANDIDATE RANKS. TB AND TD ARE THE TWO CHANNEL GENERATORS DESCRIBED IN SECTION V-A. “ $\mathcal{U}$ ” MEANS THE SNR IS DRAWN PER TRIAL AND BINNED AFTERWARDS.

Config.	T	Order search	n_{cz}	P	RSR (dB)	Gen.	SNR (dB)
A	100	25	4	10, 30	12	TB	$\{-5, \dots, 20\}$
B	50	20	4	30	12	TB	$\{-5, \dots, 20\}$
C	100	25	4	20	10	TD	$\mathcal{U}[-10, 20]$
D	50	20	4	30	12	TB	5

HS-GS performs better than EM-GS at all six SNR values. The improvement ranges from 2.46 to 3.05 dB, with an average improvement of 2.68 dB. The Cadzow step is active in all trials for this experiment.

The EM-GS result is also close to the ordinary CRLB at SNR values of 5 dB and above, with a difference of at most 0.143 dB. This shows that the ordinary CRLB gives a useful reference for the baseline estimator.

The CCRB is around 7.05–7.11 dB below the ordinary CRLB. Recall that the main difference between these two references is that the CCRB uses the known geometric structure of the channel. The gap between them therefore shows the possible benefit of using this additional channel information. HS-GS uses some of this structure and improves over EM-GS, but it still remains 4.39–4.93 dB above the CCRB.

We define the HS-GS gain as

$$\Delta_{\text{HS}} = \text{NMSE}_{\text{EM-GS}} - \text{NMSE}_{\text{HS-GS}}, \quad (44)$$

where a positive value means that HS-GS performs better than EM-GS.

C. Effect of Array Size

We next study how the number of array elements affects the benefit of the Hankel structure. We test

$$N \in \{8, 16, 32\}. \quad (45)$$

For each array size, the result is averaged over two pilot counts and six SNR values. This gives a total of 12 operating points. The results are summarized below.

N	Ratio of sums	Paired median	Baseline NMSE
8	−0.254	+0.225	−8.705
16	+0.966	+1.081	−8.672
32	+2.934	+3.226	−8.700

The benefit of HS-GS becomes larger as the array size increases. Using the paired median, the gain increases from 0.225 dB when $N = 8$ to 3.226 dB when $N = 32$.

This can be understood using the maximum Hankel rank discussed earlier, (18). For example,

$$\begin{array}{c|ccc} N & 8 & 16 & 32 \\ \hline r_{\max} & 4 & 8 & 16 \end{array} \quad (46)$$

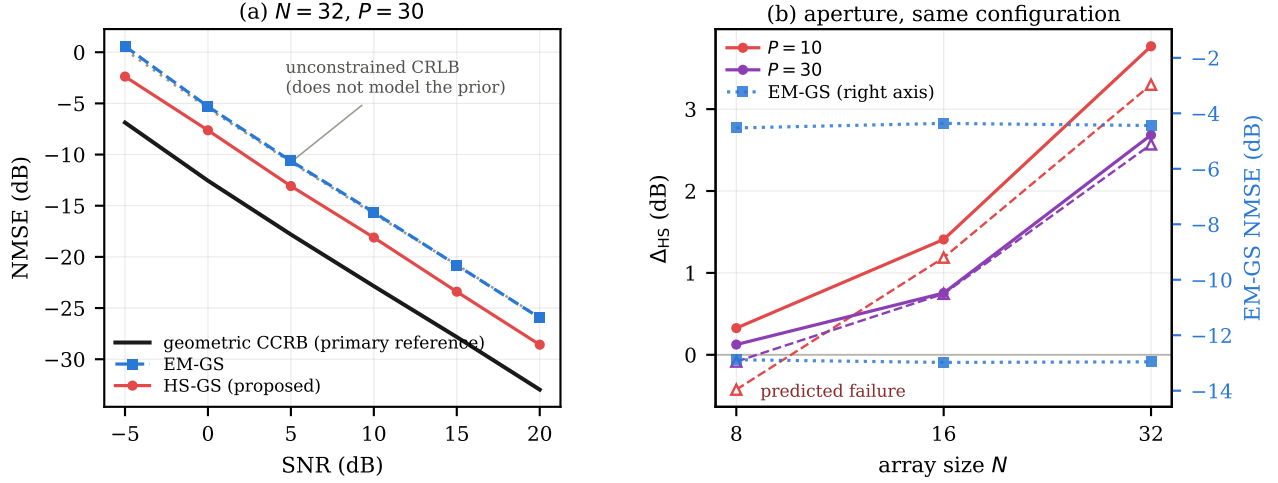


Fig. 2. Performance of EM-GS and HS-GS using configuration A. (a) NMSE versus SNR for $N = 32$ and $P = 30$, together with the CRLB and CCRB reference curves. (b) HS-GS gain versus the number of array elements for two pilot counts. Positive gain means that HS-GS performs better than EM-GS.

When $N = 8$, the Hankel matrix can have a maximum rank of only 4. If the channel contains several important components, there may not be much low-rank structure left to use. Forcing the estimate to a smaller rank can then remove part of the actual channel.

When $N = 32$, the maximum Hankel rank increases to 16. The same type of multipath channel can now use a smaller part of the available rank. This means there is more useful low-rank structure for HS-GS to exploit, which explains why the improvement becomes much larger as N increases.

An important observation is that EM-GS itself does not improve much when the array size increases. Its NMSE changes by less than 0.2 dB between $N = 8$ and $N = 32$. Without the Hankel constraint, the channel coefficients at different array elements are treated separately. Adding more array elements therefore gives more measurements, but it also adds more unknown channel coefficients.

At $N = 8$, the two ways of measuring the HS-GS improvement give different answers. The paired median gives a small positive gain of 0.225 dB, while the ratio of the total squared errors gives -0.254 dB. A negative value means that HS-GS has a larger total squared error than EM-GS. The open triangle in Fig. 2(b) marks this point: we had registered in advance that the method would fail at the smallest array, with a predicted ratio-of-sums value of -0.20 dB in a band $[-0.35, +0.05]$, and the measured -0.254 dB falls inside it.

That prediction was only half right, and the half that was wrong is the more interesting one. Three of the twelve $N = 8$ operating points came out *positive* with intervals excluding zero. So the failure at $N = 8$ is not uniform across conditions: it is an aggregate result that hides operating points where the constraint helps.

This happens partly because HS-GS does not apply a low-rank reduction in every trial. At $N = 8$, the step is inactive in 41.8% of the trials, so HS-GS gives exactly the same result as EM-GS in those trials. This keeps the typical, or median, difference close to zero. In some of the remaining trials, however, the small Hankel rank can cause the step to

remove useful channel components. These larger errors have more effect on the total squared error, which explains why the two performance measures give different results at $N = 8$.

We report both numbers rather than choosing one. Neither is wrong: they answer different questions. The paired median describes what happens in a typical trial, and the ratio of sums describes what happens to the total squared error, which a few bad trials can dominate. Because they disagree here, a single summary would hide the behaviour, and the fallback to EM-GS should not be read as a guarantee that HS-GS can never do harm.

D. Effect of Channel Complexity

We next study whether the number of propagation paths affects the benefit of the Hankel constraint. We keep the array size fixed at $N = 32$ and the SNR at 5 dB, and change only the number of paths L . The same value of L is used for all users. As described in (12), the total received power is kept fixed while L changes, so only the complexity of the channel varies.

The results show a clear trend. When the channel has only $L = 2$ paths, HS-GS improves over EM-GS by 7.04 dB. As the number of paths increases, the improvement becomes smaller. At $L = 16$, the gain is -0.12 dB, which means that the Hankel constraint no longer helps.

This can be understood from the rank of the Hankel matrix. Since $N = 32$, the maximum Hankel rank is

$$r_{\max} = \left\lfloor \frac{32}{2} \right\rfloor = 16. \quad (47)$$

Consider the two ends of the experiment. With only two paths, the channel can have a Hankel rank of at most 2, while the matrix can support a rank of up to 16. The channel is therefore strongly low rank, and there is useful structure for HS-GS to use. In contrast, when $L = 16$, the channel can use most or all of the available Hankel rank. In this case, forcing a low-rank approximation can remove an important part of the real channel instead of only removing estimation error.

The EM-GS baseline changes by only 0.191 dB over the same experiment. Therefore, the large change in HS-GS gain mainly comes from the changing usefulness of the Hankel constraint.

These results show that the Hankel constraint works best when the channel has relatively few important components compared with the maximum Hankel rank. However, simply counting the number of paths is not always enough to describe the real complexity of the channel. We take a closer look at this next using effective rank.

E. Effective Rank

Two channels can have the same number of paths but still have different levels of complexity. For example, consider two channels that both have four paths. In the first channel, all four paths may have similar strength. In the second channel, two paths may be strong while the other two are very weak. Although both channels have $L = 4$, the second channel can behave more like a channel with only two important components.

The singular values of the Hankel matrix help us measure this difference. A large singular value represents an important component of the matrix, while a very small singular value represents a weak component. For example, consider

$$[10, 9, 8, 7] \quad (48)$$

and

$$[10, 8, 0.2, 0.1]. \quad (49)$$

Both matrices can have rank 4. However, in the second case, most of the matrix can be explained by the first two components. Its real complexity is therefore much smaller than its ordinary rank suggests.

To measure this, we use the effective rank proposed in [13],

$$r_{\text{eff}} = \exp \left(- \sum_i p_i \ln p_i \right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \quad (50)$$

where σ_i are the singular values of the Hankel matrix. The value p_i represents the relative importance of each singular value. If many singular values are important, r_{eff} is large. If only a few are important, r_{eff} is small.

We then compare the effective rank with the highest rank that the Hankel matrix can support. We define

$$\rho = \frac{r_{\text{eff}}}{r_{\text{max}}}. \quad (51)$$

The value ρ can be understood as the fraction of the available Hankel rank that is effectively being used by the channel. For example, if

$$r_{\text{eff}} = 4, \quad r_{\text{max}} = 16, \quad (52)$$

then

$$\rho = \frac{4}{16} = 0.25. \quad (53)$$

This means that the channel effectively uses about one quarter of the available Hankel rank. We would expect the low-rank structure to be useful in this case. If instead $r_{\text{eff}} = 14$, then

$$\rho = \frac{14}{16} = 0.875. \quad (54)$$

The channel now uses most of the available rank, so there is much less low-rank structure left to use.

Fig. 3 shows the HS-GS gain against ρ for $N \in \{16, 32, 64\}$. The gain approaches zero at

$$\rho = 0.588, \quad 0.518, \quad 0.544, \quad (55)$$

for $N = 16, 32$, and 64 , respectively.

Three things must be said about those three numbers before anything is concluded from them, and each one weakens the reading.

Only the middle one is measured on both sides of zero. At $N = 32$ the sweep contains a cell with a positive gain and a cell with a negative gain, so the crossing lies between two measurements. At $N = 16$ and $N = 64$ every measured cell has a positive gain, so those two crossings are obtained by continuing a line past the last measured point. They are extrapolations, and an extrapolation is a weaker number than an interpolation.

The three are not the same statistic. The $N = 32$ value comes from a sweep at a fixed SNR of 5 dB. The other two come from cells in which the SNR is drawn over the whole range and the result is pooled, so their value depends on how many low-SNR trials happened to be drawn. Comparing a fixed-SNR number with two pooled ones is not a like-for-like comparison.

The curves do not lie on top of each other. Comparing $N = 16$ with $N = 64$ directly — the one comparison here that is like-for-like, since those two share a design, a draw and an estimator — the $N = 16$ curve sits above the $N = 64$ curve at every matched value of ρ , by 0.49 dB on average and by 0.90 dB at worst. A one-signed gap of half a decibel is a systematic offset, not agreement.

What survives is weaker than an invariance and still useful: across a factor of four in array size, the gain turns from clearly positive to near zero somewhere in the region of $\rho \approx 0.5$ – 0.6 , and the ordering of the curves does not change. In the cases tested here, the Hankel constraint is most useful when the channel effectively uses about half or less of the available Hankel rank.

We also carried out a test to see what happens when the channel becomes more complex and the low-rank constraint starts to become harmful. In this test the Cadzow step was forced to remain active, so the estimator was denied the option of declining to constrain. This is a diagnostic and not the proposed method. At $\rho = 0.735$, the gain becomes -0.075 dB with a 95% interval of $[-0.186, -0.026]$ dB, which excludes zero. Forcing a low-rank constraint on a complex channel therefore does measurable harm, not merely no good.

At that same operating point the unforced estimator declines to constrain on 38.8% of trials. Selecting the full rank is what protects it, and on this evidence that protection is worth having. It remains a property of the operator rather than a property of the selection rule: these measurements show that abstention is available and that it is used often here, not that it is chosen whenever it should be.

1) Is ρ actually the right descriptor?: Up to this point we have shown that ρ organises the results. We have not shown that it organises them better than something simpler, and those

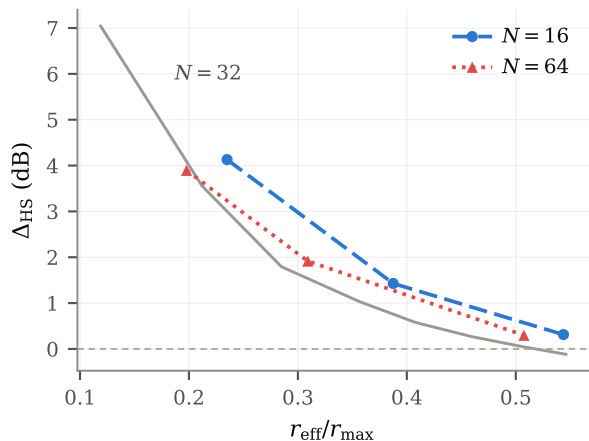


Fig. 3. HS-GS gain against the normalized effective rank $\rho = r_{\text{eff}}/r_{\text{max}}$ for $N \in \{16, 32, 64\}$. The gain approaches zero in a similar region of ρ for the three array sizes. Only the $N = 32$ crossing is bracketed by measurements of both signs; the other two are extrapolated past the last measured cell, and the $N = 16$ curve sits above the $N = 64$ curve at every matched ρ by 0.49 dB on average.

are different claims. An earlier version of this paper made the second claim without testing it. This subsection tests it.

We compare three candidate descriptors, each used in exactly the same way: the path count L , the normalized path count L/r_{max} , and $\rho = r_{\text{eff}}/r_{\text{max}}$. The prediction rule is piecewise-linear interpolation on a fixed eight-point table, with no regression and no curve fitting. The table is the $N = 32$ path-count sweep, and the same eight rows are used for all three descriptors; only the horizontal coordinate changes. We then predict six cells at $N = 16$ and $N = 64$ that are not in the table, and compare against what was measured there.

Descriptor	Mean error (dB)	Worst (dB)	Sign correct
L	1.91	2.85	5 of 6
L/r_{max}	0.34	1.06	6 of 6
$\rho = r_{\text{eff}}/r_{\text{max}}$	0.46	1.61	5 of 6

The result is clear in one direction and negative in the other.

Normalizing by array capacity is what matters. Raw path count is a poor predictor across array sizes, with a mean error of 1.91 dB, and it could not be otherwise: $L = 7$ is close to the ceiling when $r_{\text{max}} = 8$ and far below it when $r_{\text{max}} = 32$, so the same L describes opposite situations. Dividing by r_{max} removes that problem and cuts the error by more than a factor of five.

The effective rank buys nothing beyond that. ρ is not better than the much simpler L/r_{max} ; it is slightly worse on both mean error and sign agreement. The difference between them is not resolved by six cells — the paired interval on the difference is $[-0.16, +0.40]$ dB and includes zero — so the honest statement is that this evidence does not separate them, and certainly does not favour ρ . Repeating the comparison on the wider $\text{SNR} \geq 5$ dB summary gives the same ordering, 0.56 dB for L/r_{max} against 0.61 dB for ρ .

We therefore do not claim that effective rank describes this behaviour better than path count does. What the experiments support is narrower: *path count normalized by the array's*

structural capacity predicts when the constraint helps, and ρ is one way of performing that normalization which this evidence does not distinguish from the simpler one. We continue to use ρ in this paper because it extends naturally to channels whose paths have unequal strengths, where L is ambiguous — but that is a reason of convenience, and we report it as one.

Two limits apply to ρ whichever way it is read. It does not determine the size of the improvement, since the gain still changes with array size at matched ρ . And it is computed from the true noiseless channel, which the receiver does not have, so it explains the simulation results rather than serving as a quantity an estimator could evaluate before estimating.

F. Evaluation on Different Channel Distributions

So far, the relationship between ρ and the HS-GS gain has been obtained from our original channel simulations. We now test whether this relationship is still useful when the paths are generated differently.

The idea of this test is simple. We first use the relationship already obtained between ρ and the HS-GS gain to make a prediction. We then generate channels using a different distribution, run HS-GS, and compare the measured gain with the prediction. The previously obtained relationship is not changed using the new results.

For example, the first test uses a clustered Saleh–Valenzuela channel. In this model, propagation paths occur in clusters instead of being generated in the same way as in the original simulations. For this channel distribution, the normalized effective rank is

$$\rho = 0.331. \quad (56)$$

From the previously obtained relationship, this value of ρ predicts an HS-GS gain of

$$\Delta_{\text{HS}} = 1.30 \text{ dB}. \quad (57)$$

After running 276 trials, the measured gain is 1.173 dB, with a 95% confidence interval of $[1.017, 1.380]$ dB. Therefore, the difference between the predicted and measured gains is only 0.13 dB.

A second channel distribution is tested in the same way. This one uses a fixed number of $L = 10$ paths for every user, with gains drawn as $\alpha_\ell \sim \mathcal{CN}(0, 1)$ and angles drawn uniformly on $(-90^\circ, 90^\circ)$, following the receive-side channel configuration of [14]. It is therefore quite different from the first test: the clustered model groups its paths together, while this one spreads ten independent paths across the whole angular range. In this case,

$$\rho = 0.400. \quad (58)$$

The predicted gain is 0.648 dB. After 266 trials, the measured gain is 0.963 dB, with a 95% confidence interval of $[0.808, 1.086]$ dB. The prediction error is 0.315 dB.

The two results are summarized below.

ρ	Predicted gain	Measured gain	Error
0.331	1.30 dB	1.173 dB	0.13 dB
0.400	0.648 dB	0.963 dB	0.315 dB

The second prediction is noticeably less accurate than the first, and it falls outside the confidence interval of the

measured gain. That interval describes the uncertainty of the measurement and not the uncertainty of the prediction, so this is not a formal rejection of the relationship. It does mean we should not describe the agreement as close in both cases. The honest summary is that the relationship places an unseen channel distribution on the correct part of the curve, and that it is not accurate to a tenth of a decibel in general.

These results are important because the new channel distributions were not used to obtain the original relationship between ρ and the HS-GS gain. The relationship still gives reasonably close predictions when the path distributions are changed.

However, these tests do not use a completely different channel model. Both new channel distributions still follow the geometric channel model in (1). What changes is how the path numbers, path strengths, and path angles are generated. Therefore, these results show that the relationship can extend across different multipath distributions within the channel model considered in this work.

G. Additional Results

We also look at how the number of users affects the results and whether it has a large effect on the benefit of HS-GS. We test

$$K \in \{2, 3, 4\}. \quad (59)$$

For SNR values of 5 dB and above, the HS-GS gain changes by only 0.094 dB across these three cases. Therefore, within the range tested, changing the number of users has only a small effect on the improvement provided by the Hankel constraint.

We also examine the computational cost of HS-GS. The main extra cost comes from choosing the Hankel rank. This rank-selection step accounts for 57.7–85.5% of the total run-time.

Finally, we test whether the Cadzow step needs to be applied repeatedly during the EM-GS iterations. We compare two approaches. In the first approach, the step is applied during the EM-GS iterations. In the second approach, EM-GS is completed first and the Cadzow step is applied only once at the end.

The difference in mean gain between these two approaches is only 0.003 dB. This is a very small difference. It suggests that, in the cases tested here, the biggest benefit comes from applying the Hankel structure itself rather than from applying the step repeatedly during the EM-GS iterations. Since the one-shot version is also cheaper, this is a useful practical result, although a small difference in the mean does not prove that the two are equivalent at every operating point.

VI. LIMITATIONS AND CONCLUSION

This work studies when low-rank Hankel structure can help magnitude-only channel estimation for Rydberg atomic receivers.

The main point is easy to understand. The Hankel constraint is useful when the channel is simple compared with the rank that the Hankel matrix can support. For example, a channel that only needs a few important components has more low-rank

structure that can be used during estimation. As the channel becomes more complex and uses most of the available rank, the benefit becomes smaller. If a low-rank constraint is forced on a channel that is too complex, it can eventually make the estimate worse.

The experiments first show this using the number of propagation paths. At $N = 32$, the HS-GS gain decreases from 7.04 dB when $L = 2$ to -0.12 dB when $L = 16$. However, the number of paths alone is not a complete measure of channel complexity because some paths can be much weaker than others.

What the experiments support is that path count must be measured *relative to the array's structural capacity*. Raw path count does not transfer between array sizes, because $L = 7$ is near the ceiling when $r_{\max} = 8$ and far below it when $r_{\max} = 32$. In a held-out test on six cells at $N = 16$ and $N = 64$, predicting from raw L gives a mean error of 1.91 dB, while predicting from the normalized count $L/r_{\max}$ gives 0.34 dB.

We use the normalized effective rank

$$\rho = \frac{r_{\text{eff}}}{r_{\max}} \quad (60)$$

as our normalization throughout, because it extends to channels whose paths have unequal strengths, where counting paths is ambiguous. On the same held-out test it scores 0.46 dB, which is not better than the simpler $L/r_{\max}$ and not separated from it by six cells. **We therefore do not claim that effective rank describes this behaviour better than path count does;** the claim we make is about normalizing by capacity, and ρ is one way to do it.

For $N = 16, 32$, and 64 , the HS-GS gain approaches zero at $\rho = 0.588, 0.518$, and 0.544 , so a factor of four in array size leaves the crossing in the same region. Two of those three values are extrapolated rather than bracketed by measurements of both signs, and the $N = 16$ curve sits about 0.49 dB above the $N = 64$ curve at every matched ρ , so this is a similar region and not a coincident one. In the cases tested here, the Hankel constraint is most useful when the channel effectively uses about half or less of the available Hankel rank.

The relationship is also examined using two additional channel distributions that were not used to build it. The predicted HS-GS gains differ from the measured gains by 0.13 and 0.315 dB.

Limitations. All results in this work are obtained from simulations. We have not yet tested the method using measured hardware data or recorded Rydberg channels. The additional channel distributions also still follow the geometric ULA model in (1). Therefore, the results do not show that the same behaviour will hold when the actual channel does not follow this model.

The results also apply only to the magnitude-only Rydberg architecture described in Section II. A superheterodyne Rydberg receiver recovers both amplitude and phase, and the problem studied here does not arise for it.

Another limitation is that ρ is calculated using the true noiseless channel in our simulations. The receiver does not know this channel in practice. Therefore, ρ is currently used to explain when the method works rather than as a value that the receiver can calculate before channel estimation.

The Cadzow step used here is a denoising step and not a projection, as shown in Section III-B. It is also a weaker requirement than the array model itself, so a result about this constraint is not automatically a result about the physical multipath model.

Finally, although the point where the HS-GS gain approaches zero is similar across the tested array sizes, the exact amount of gain does not perfectly follow ρ . Therefore, ρ helps explain the useful region of the Hankel constraint, but it does not completely predict its performance in every case. The CRLB and CCRB are also used only as theoretical reference curves, because the estimators considered here are not assumed to be unbiased and because we do not give enough detail for those curves to be independently reproduced.

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